

Q7

$$S = \{0, 1, 2\}$$

$$P = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & 0 & \frac{1}{2} \end{pmatrix}$$

$$P^2 = \begin{pmatrix} \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{4} & \frac{1}{4} \end{pmatrix}$$

$$P(X_2 = 0, X_4 = 0) = \cancel{P(X_2 = 0) \cdot P(X_4 = 0)}$$

$$= P(X_2 = 0) \cdot P(X_4 = 0 | X_2 = 0) =$$

$$= P(X_2 = 0) \times \frac{1}{4}$$

$$P(X_2 = 0) = \sum_{i=0}^2 P(X_0 = i) \cdot p_{i0}^{(2)} = \frac{1}{3} \left(\frac{1}{4} + \frac{1}{4} + \frac{1}{2} \right) =$$

$$= \frac{1}{3} \times \frac{1}{4} = \frac{1}{12}$$

2. a)

• Set $\{1, 2\}$.

States 1 and 2 communicate
From $\{1, 2\}$ it is impossible to
reach any other states

$C_1 = \{1, 2\}$ is a closed irreducible
set of recurrent states.

Set $\{3, 4\}$

$3 \rightarrow 4$ ($p_{34} = 0,75$)

$4 \rightarrow 3$ ($p_{43} = 0,75$)

No transitions out of $\{3, 4\} \rightarrow$ closed

$3 \leftrightarrow 4 \rightarrow$ irreducible

closed + irreducible + finite $\rightarrow$ recurrent

Set $\{5, 6\}$

S: $p_{51} = 0,5$, $p_{54} = 0,3 \rightarrow$ can reach C_1, C_2

G: $p_{62} = 0,2$, $p_{63} = 0,2$, $p_{64} = 0,2 \rightarrow$ can reach
 C_1 and C_2

Once in C_1 or C_2 cannot return to S or G

transient

$$S = \{5, 6\} \cup \{1, 2\} \cup \{3, 4\}$$

2 b)

Within Recurrent Classes

$$f_{11} = f_{12} = f_{21} = f_{22} = 1$$

$$f_{33} = f_{34} = f_{43} = f_{44} = 1$$

Between Closed Classes / Recurrent $\rightarrow$ Transient

$$f_{ij} = 0 \text{ for } i \in \{1, 2\}, j \notin \{1, 2\}$$

$$f_{ij} = 0 \text{ for } i \in \{3, 4\}, j \notin \{3, 4\}$$

Transient $\rightarrow$ Recurrent

to C_1 :

$$f_{51} = 0,5 + 0,1f_{51} + 0,1f_{61}$$

$$f_{61} = 0,2 + 0,2f_{51} + 0,2f_{61}$$

$$f_{51} = f_{52} = \frac{3}{5}, \quad f_{61} = f_{62} = \frac{2}{5}$$

to C_2 :

$$f_{53} = 0,3 + 0,1f_{53} + 0,1f_{63}$$

$$f_{63} = 0,4 + 0,2f_{53} + 0,2f_{63}$$

$$f_{53} = f_{54} = \frac{2}{5}, \quad f_{63} = f_{64} = \frac{3}{5}$$

transient $\rightarrow$ transient

$$f_{56} = 0,1 + 0,1 f_{56} \rightarrow f_{56} = \frac{1}{9}$$

$$f_{65} = \frac{1}{4}$$

$$f_{55} = \frac{1}{8}$$

$$f_{66} = \frac{2}{9}$$

$$\begin{array}{c|cccccc} 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ 2 & 1 & 1 & 0 & 0 & 0 & 0 \\ 3 & 0 & 0 & 1 & 1 & 0 & 0 \\ 4 & 0 & 0 & 1 & 1 & 0 & 0 \\ 5 & \frac{3}{5} & \frac{3}{5} & \frac{2}{5} & \frac{2}{5} & \frac{1}{8} & \frac{1}{9} \\ 6 & \frac{2}{5} & \frac{2}{5} & \frac{1}{5} & \frac{1}{5} & \frac{1}{4} & \frac{2}{9} \end{array}$$

3.

$$|S| = 52!$$

$$\pi(\sigma) = \frac{1}{52!} \quad \forall \sigma \in S_{52}$$

$$E(\tau_\sigma) = \frac{1}{\pi(\sigma)} = 52!$$

$$E(\tau) = \frac{52!}{60 \times 60 \times 24 \times 365} \approx 2,56 \times 10^{60} \text{ years}$$

Q4

$$p_0 = 0,5$$

$$p_1 = 0,1$$

$$p_2 = 0$$

$$p_3 = 0,4$$

$$\mu = 0(0,5) + 1(0,1) + 2(0) + 3(0,4) = 1,3$$

$\mu > 1$ extinction not certain

$$\text{PGF: } G(s) = 0,5 + 0,1s + 0,4s^3$$

$$q = 0,5 + 0,1q + 0,4q^3$$

$$4q^3 - 9q + 5 = 0 \quad (*)$$

$$(q-1)(4q^2 + 4q - 5) = 0 \quad (**)$$

$$q = \frac{-4 \pm \sqrt{16 + 80}}{8} = \frac{-1 \pm \sqrt{6}}{2}$$

Smallest non-negative root